

comes with on-camera analytics and features SuperResolution mode that quadruples the resolution into 640 × 480 or even 1024 × 768, which is full HD (FHD).

FLIR has come out with a thermal imager (TIK) that comes with a sensor having a 1024 × 768 HD detector and features up to 3.1 megapixels of detection with its UltraMax processing technology. It also comes with HDIR lenses that claim to deliver more accurate measurements from a greater distance. (FLIR claims twice the distance as legacy systems.) Additionally, it features video recording along with continuous auto-focus. FLIR also packs in some software tools that allow users to analyse data, change colour patterns and generate detailed reports.

Keysight U5850 series is also claimed to be among the first to come with four times in-camera fine resolution of 320 × 240 pixels from a 160 × 120 pixel detector that comes with manual-focus.

IR thermometer ILV-301 by Flex Instruments features a distance-to-spot ratio of 30:1 calculated at 95 per cent energy. It can sense a range from -30°C to 800°C.

MECO has a thermometer called IRT1050P that can sense from -50°C to 1050°C. It features a distance-to-spot ratio of 50:1, too.

Flex Instruments' thermal camera (model number TG-501) features a 128 × 128 pixel thermal image after interpolation. Its display is a 160 × 128 pixel TFT-LCD, mounted on a pistol grip with one navigator key operation.



FLIR T1020

Bird's-eye view of the trends

1. Decrease in product price due to availability of lower-resolution devices. Vendors have introduced many entry-level devices with lower feature sets that enabled them to bring down the price
2. Decrease in product cost due to sensor manufacturers moving from 15.2cm (6-inch) to 20.3cm (8-inch) production lines
3. Many new products with better protection from environmental influences like rain, fog and sunlight as well as the capability to survive a roughly 60cm drop
4. Better image-processing solutions in addition to higher resolutions
5. Laser focus system introduced in thermal imagers
6. Wi-Fi connectivity for data transfer through smartphones

How the ease of use of thermal imagers improved over the years

Many things got developed over the years as listed below:

1. Movable optics: One of the major problem a user used to face in the field was pain in the neck while continuously using the camera, especially in the case of harsh environmental conditions. For such applications, FLIR rolled out a movable/rotating optics design, which was accepted globally with high enthusiasm. This design eases the operation with almost no pressure on the neck while using the unit for looking at hotspots on electrical installations or places that are not easy to see.
2. PDF report generation in the field by camera: The same can even be transmitted over Wi-Fi using smartphones in the field. This reduces the report time to a few minutes and brings in efficiency.
3. Quality of image: As very high NETD levels of nearly 20mK are available in uncooled cameras at 30Hz frequency, the images are very crisp and clear, resulting in excellent image quality. This helps to pin-point problematic areas quickly.
4. JPEG format: Images taken in the field can also be viewed on mobiles or PCs that do not have dedicated software installed. Images can be stored in JPEG format and can be viewed even with basic picture-viewing software.
5. Complete range: The lowest-range camera starts at ₹ 45,000, empowering many officials who have tight budgets.

—T.P. Singh, country manager, India-Instruments, FLIR Systems India Pvt Ltd

Making it easier to use thermal cameras

FLIR TG165 was announced late last year, which is an easy-to-use thermal imager built for do-it-yourself users and for industrial professionals who require a compact low-cost tool for quick troubleshooting.

Last year, Fluke talked about its introduction of a laser focus system in thermal imagers where a laser beam was used to measure the distance, which aids the user to focus properly on the object.

FLIR TG165 comes with dual rotating lasers to visually mark the size and location of what is being measured, while an on-screen crosshair pinpoints

the region on the thermal image.

FLIR C2 was also launched last year as a pocket-size camera designed to show hidden heat patterns due to energy leaks, bad wiring, HVAC issues and other problems. FLIR's TIK also comes with a rotating block that houses the optics so you can use it on tougher angles while holding your head in different positions when you are using the touchscreen to operate the device.

Start debugging those electronics

You can now pick up a thermal imager and start working on debugging your circuit board. It will help you find that one overheating chip in the board that is causing the embedded system to fail, or quickly identify that short in the circuit. ●